

1.6 Classwork: Sampling bias and survey methodologies — Markscheme

Calculator allowed. Marks are not printed on the student sheet; they are recorded here. Total: 17 marks.

Throughout: a technique must be named with its IB term (simple random, convenience, systematic, quota, stratified). An explanation earns its R1 only when it is tied to the context of the question, not a general definition alone.

1. [Maximum mark: 8]

Grades 9–12: 84, 78, 68, 70 students; total 300. Sample size 30.

(a) All the students at the high school (300 students). **A1**

“The 30 students surveyed” or “Grade 12 students” earns no mark — that is a sample, not the population.

(b) Convenience sampling **A1**

Only Grade 12 students are surveyed, and only those who choose the common room, so the sample is not representative of all grades; older students may have different sleep habits. **R1**

Accept any reason tied to the context: only one grade is included; friends of Alex or students with the same routine are over-represented; not every student has an equal chance of being chosen.

“The sample is too small” earns no R1 — the sample size is the same 30 as the later stratified sample.

(c)(i) $\frac{78}{300} \times 30$ **M1**

$= 7.8$ **A1**

≈ 8 students **AG**

Accept $\frac{30}{300} = \frac{1}{10}$ of each grade, so $\frac{78}{10} = 7.8$. The value 7.8 must be seen for the A1; writing only “8” earns the M1 at most.

(c)(ii) Grade 9: $8.4 \approx 8$; Grade 11: $6.8 \approx 7$; Grade 12: $7.0 = 7$ **A1A1**

A1 for two correct, A1 for all three correct. Check: $8 + 8 + 7 + 7 = 30$.

Numbers equal in each grade (7.5 or “7 or 8 each”) earn no marks.

(d) Systematic sampling **A1**

Accept “systematic random sampling”. “Random sampling” alone earns no mark.

2. [Maximum mark: 9]

(a)(i) Discrete **A1**

(a)(ii) Continuous **A1**

Time is measured, not counted, so it is continuous even when it is recorded to the nearest hour or half-hour.

(b) Quota sampling **A1**

“Convenience” earns no mark: the fixed number from each group is what makes it quota sampling.

(c) In quota sampling the students within each group are not chosen at random (they are the first to arrive); in stratified sampling they are chosen at random, and the group sizes are in proportion to the population. **R1**

Award R1 for either the “not random within each group” point or the “not in proportion to the population” point.

(d) Use a random number generator (GDC or spreadsheet) to produce random whole numbers from 1 to 300 **M1**

select the students with those numbers, ignoring repeats, until 30 different students are chosen **A1**

Accept drawing 30 numbers from a hat containing 1 to 300. M1 for a random method applied to the numbered list; A1 for 30 distinct students (no repeats). “Pick 30 names from the list” with no random method earns no marks.

GDC: *randInt(1,300) repeated, or randSamp on the list 1 to 300 without replacement.*

(e) Check the value with the student, or remove it from the data set **A1**

19 hours of sleep in one night is unrealistic, so the value is most likely a recording error (for example 9 hours) **R1**

Accept “it is an outlier and is probably an error”. The R1 needs the reason that the value is implausible or an error; “it is an outlier” alone earns the A1 only. “Keep it” earns A0, even with the reason that some outliers are valid data.

(f) It is a leading question: its wording suggests the answer “yes”, so students may agree rather than give their honest opinion. **R1**

Accept “biased wording”, “it pushes students towards one answer”. “Students may lie” alone earns no mark — the R1 is for identifying that the question itself causes the bias.